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X-RAY POINT SOURCES IN THE SOMBRERO GALAXY: VERY SOFT SOURCES, THE GLOBULAR CLUSTER/LOW-MASS X-RAY BINARY CONNECTION, AND AN OVERVIEW

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ABSTRACT

We report on the population of point sources discovered during an 18.5 ks *Chandra* ACIS-S observation of the Sombrero galaxy. We present the luminosity function and the spectra of the six brightest sources, consider correlations with globular clusters (GCs) and with planetary nebulae, and study the galaxy's population of very soft sources. We detected 122 sources. Twenty-two sources are identified as very soft; of these, five appear to be classical luminous supersoft X-ray sources (SSSs), while 17 may belong to the slightly harder class referred to as quasi-soft (QSSs). There is an overdensity of very soft sources within 2 kpc of the nucleus, which is itself the brightest X-ray source. Very soft sources are also found in the disk and halo, with one QSS in a globular cluster (GC). This source is somewhat harder than most SSSs; the energy distribution of its photons is consistent with what is expected from an accreting intermediate-mass black hole. Several sources in the Sombrero's halo are good candidates for SSS models in which the accretor is a nuclear-burning white dwarf. In total, 32 X-ray sources are associated with GCs. The majority of sources with luminosity greater than 10^{38} ergs s^{-1} are in GCs. These results for M104, an Sa galaxy, are similar to what has been found for elliptical galaxies and for the late-type spiral M31. We find that those optically bright GCs with X-ray sources house only the brightest X-ray sources. We find that, in common with other galaxies, there appears to be a positive connection between young (metal-rich) GCs and X-ray sources but that the brightest X-ray sources are equally likely to be in metal-poor GCs. The luminosity function of X-ray sources in GCs has a cut-off near the Eddington luminosity for a 1.4 $M_{\odot}$ object. We propose a model that can explain the trends seen in the data sets from the Sombrero and other galaxies. Thermal timescale mass transfer can occur in some of the younger clusters in which the turnoff mass is slightly greater than 0.8 $M_{\odot}$; multiplicity may play a role in some of the most massive clusters; accretion from giant stars may be the dominant mechanism in some older, less massive and less centrally concentrated clusters. Key elements of the model can be tested.

Subject headings: galaxies: individual (M104) — X-rays: binaries — X-rays: galaxies

1. INTRODUCTION

M104 (NGC 4594), the Sombrero galaxy, is one of the most remarkable sights in the sky. Beyond its aesthetic appeal, the fact that it is a bright Sa galaxy that we view nearly edge on provides insight into the organization of matter within spiral galaxies, allowing us to identify objects located out of the plane of the disk. During its second year of operations, *Chandra* conducted an 18.5 ks observation of M104. We report here on the X-ray point sources discovered during this observation.

The results are remarkable because, in this relatively short observation of a galaxy located 8.9 Mpc from us (Ford et al. 1996), *Chandra* detected 122 point X-ray sources, 22 of them identified as very soft by an algorithm (Di Stefano & Kong 2003a, 2003b) developed to search for luminous supersoft X-ray sources (SSSs) and quasisoft X-ray sources (QSSs). These observations provide the first glimpse of very soft sources located a measurable distance away from the plane of the galaxy in which they reside.

M104 is estimated to contain ~1100 globular clusters (GCs) and therefore provides a good opportunity to study the relationship between X-ray sources and GCs. Thirty-two of X-ray sources, including one QSS, are apparently associated with globular clusters (GCs). In fact, M104 is the first spiral outside the Local Group in which the connection between GCs and X-ray sources is being studied.

M104 is a LINER galaxy, and it has been conjectured that it houses an active galactic nucleus (AGN), making the nuclear region a prime target for X-ray observations. *ROSAT* High-Resolution Imager observations identified eight point-like sources, including one coincident with the nucleus, tentatively interpreted as a low-luminosity AGN (Fabbiano & Juda 1997). *BeppoSAX* observations, coupled with a short (1.7 ks) *Chandra* exposure, also support the AGN model, although a superposition of sources in the nuclear region could not be ruled out (Pellegrini et al. 2002). The diffuse emission is also being studied (W. R. Forman et al. 2003, in preparation; see Delain et al. 2001, which also includes a preliminary analysis of the point sources).

The focus of this paper is on the rich point-source population revealed by the unique angular resolution of *Chandra*. In § 2 we discuss the observations and data reduction, focusing in § 3 on an overview of the source properties: identifications with GCs and planetary nebulae, the spectra of the nuclear source and of the six brightest nonnuclear sources,

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the luminosity function, and the color-color diagram. Our main science results on SSSs and QSSs are discussed in § 4, and on the connection between GCs and low-mass X-ray binaries (LMXBs), discussed in § 5. Section 6 highlights our conclusions and discusses prospects for future observations.

2. OBSERVATIONS AND DATA REDUCTION

M104 was observed by *Chandra* on 2001 May 31 with the Advanced CCD Imaging Spectrometer (ACIS-S). The effective exposure time was about 18.5 ks. In this paper we focus only on the data taken with the S3 CCD. The nuclear region of M104 was placed near the aim point of the ACIS-S array. All data were telemetered in faint mode and were collected with a frame transfer time of 3.2 s. In order to reduce the instrumental background, only data with *ASCA* grades of 0, 2, 3, 4, and 6 were included. We ran the detection algorithm in the energy range 0.1–7 keV and visually inspected the regions around each source. We also inspected the background count rates from the S1 chip; no significant background flares were found. The data reduction and analysis was done with CIAO, version 2.3, and energy spectra were analyzed with XSPEC, version 11.2.

Discrete sources in the image were found with WAVDETECT (Freeman et al. 2002), together with an exposure map. A total of 122 sources were detected. Source count rates were determined via aperture photometry. The radius of the aperture was varied with average off-axis angle in order to match the 90% encircled energy function. The background was extracted from an annulus centered on each source. In some cases, for example, near the nucleus, we modified the extraction region to avoid nearby sources. It was also necessary to modify the extraction radius for some faint sources that are close to more luminous sources. Every extraction region was examined carefully in the image. The count rate was corrected for exposure, background variation, and instrumental point-spread function (PSF). All sources in the catalog have signal-to-noise ratio $(S/N) > 2.6$, and only 12 sources have $S/N < 3$ with the minimum number of counts equal to 7. A total of 122 sources were detected. Figure 1 shows all the detected X-ray sources overlaid on an optical Digitalized Sky Survey. A "true-color" image of the central $4' \times 4'$ region is also shown in Figure 2.

Table 1 lists the 122 sources in our catalog, sorted in order of increasing right ascension (R.A.). The position of each detected source listed in Table 1 was corrected for astrometry with the Two Micron All Sky Survey (2MASS) catalog (Cutri et al. 2000). Within S3, we found six X-ray sources located within $1''$ of 2MASS stars (see Table 1). The average uncertainty of the X-ray position is about $0''.1$. We calculated the average coordinate shift for the six sources and used this to correct the *Chandra* position. The shift in R.A. and declination is $0''.17$ and $0''.25$, respectively. The average displacement between the X-ray and 2MASS positions after the shift is about $0''.15$.

Some of the sources may be background active galactic nuclei (AGNs). We estimated the contribution of background unrelated objects by the *Chandra* Deep Field Surveys (e.g., Brandt et al. 2001; Giacconi et al. 2001). We found that at 4.5×10^{37} ergs s^{-1} (the completeness limit), fewer than 10 sources are likely to be background objects. We expect that only a small fraction of the ~10 background objects are soft enough to be identified as very soft sources.

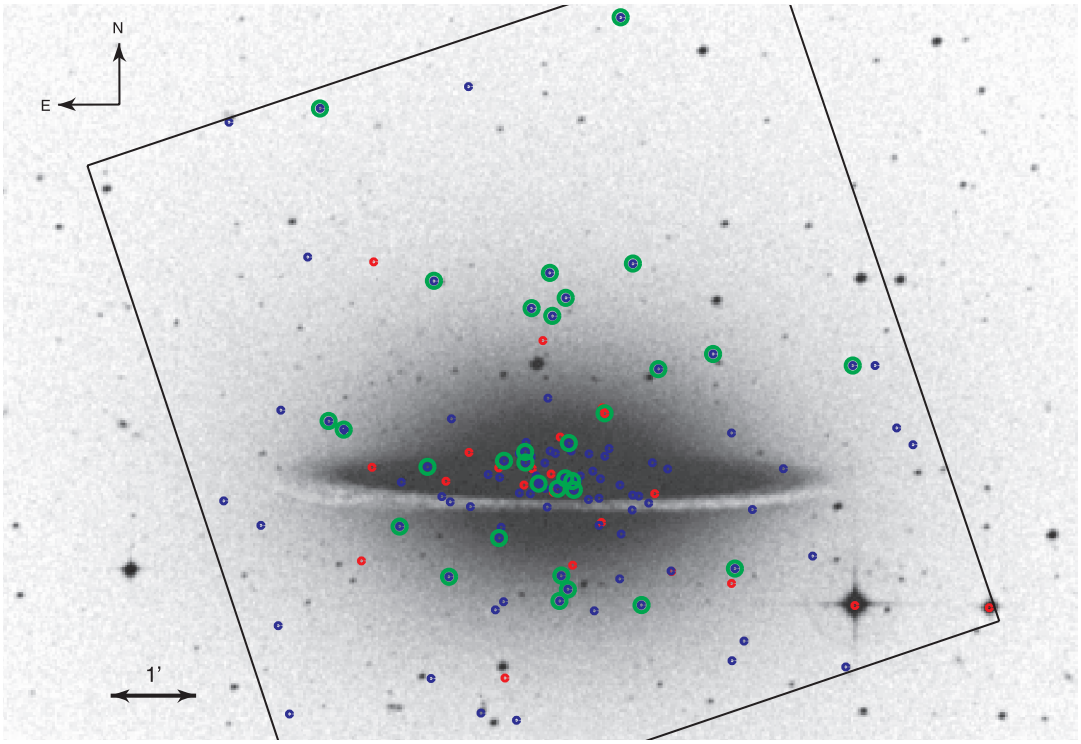


FIG. 1.—Detected X-ray sources (*blue and red filled circles*) overlaid on an optical Digitalized Sky Survey image of M104. SSSs and QSSs are shown as red filled circles, while X-ray GCs are marked with green open circles. To be identified as a match, a GC and an X-ray source must lie within $1''.5$ of each other; note, however, that to show the matches clearly, the green circles have a much larger radius ($5''$). The field of view of the S3 CCD is marked as a black line. North is up, and east is to the left.

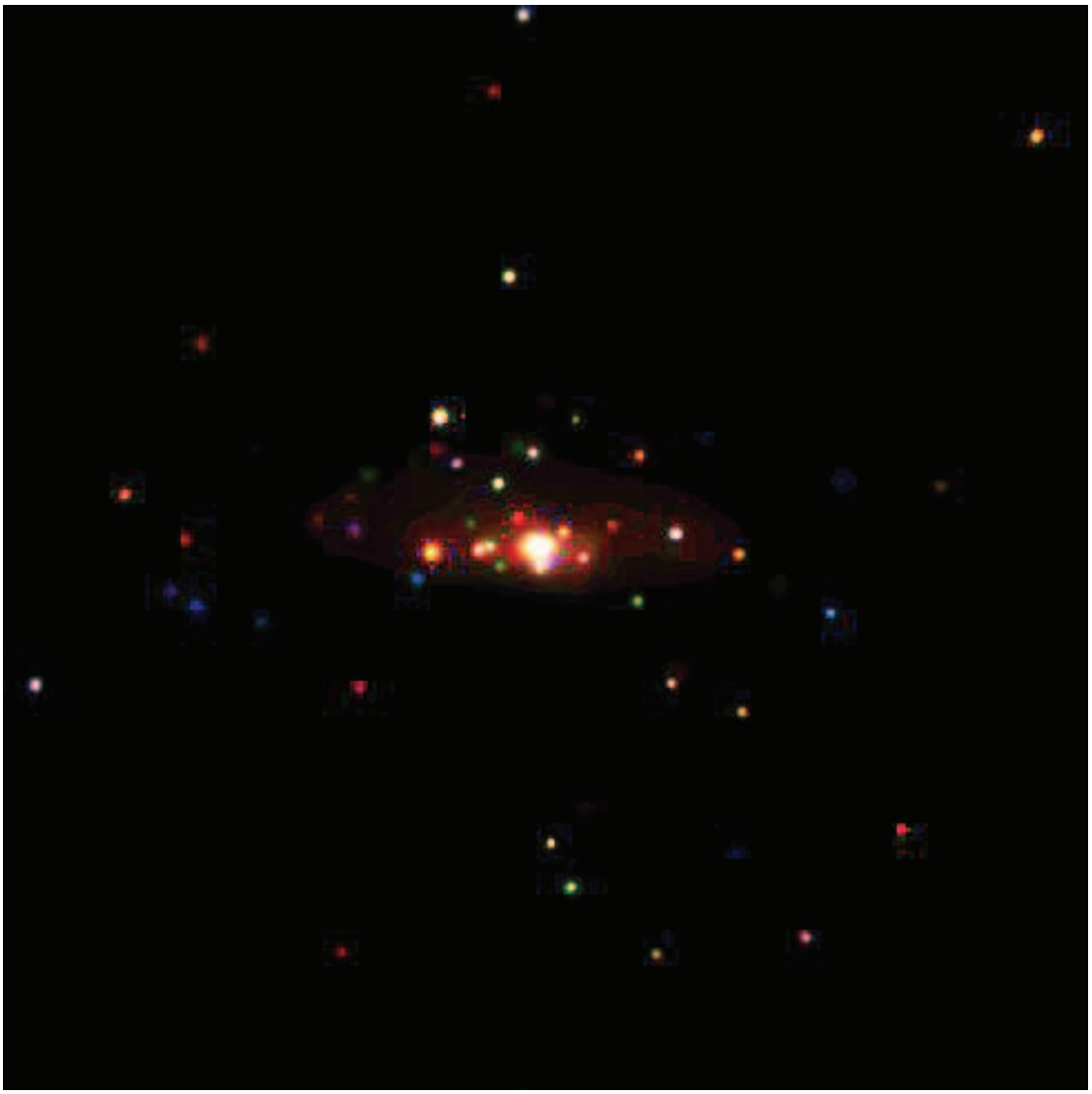


FIG. 2.—"True color" *Chandra* image of the central $4' \times 4'$ region of M104. The three energy bands are 0.1–1.1 keV (*red*), 1.1–2 keV (*green*), and 2–7 keV (*blue*). The image has been exposure-corrected and adaptively smoothed. North is up, and east is to the left. Note the extended diffuse emission near the galaxy's center. The background subtraction method that we applied should minimize the influence of this diffuse emission on estimates of the broadband count rates of point sources; nevertheless, the uncertainty in the numbers of counts in each band (S, M, and H) is larger than it would otherwise be for those X-ray sources located in regions with significant amounts of diffuse emission.

To quantify this, we have studied data from several fields observed by *Chandra*[6] and find that one to three very soft sources unrelated to M104 are likely to be present in our data; these are most likely to be foreground stars (see, e.g., Di Stefano et al. 2003). In fact, two foreground stars are identified as QSSs. For more details on how the analysis was conducted, see Di Stefano et al. (2001) and Kong et al. (2002a).

3. PROPERTIES OF X-RAY SOURCES

3.1. *Source Identification*

Observations at optical wavelengths have identified globular clusters and planetary nebulae in M104 (VanDalfsen et al. 2001; Larsen, Forbes, & Brodie 2001; Ford et al. 1996). We searched for coincidences between these catalogs and our *Chandra* sources. There are 640 GCs identified from the ground (VanDalfsen et al. 2001) in the field of view of S3. Moreover, 72 additional GCs were found by using the *Hubble Space Telescope* (*HST*; Larsen et al. 2001). It is worth noting that the astrometry of the *HST* image (Larsen et al. 2001) was not well calibrated, and we therefore corrected the astrometry by using the 2MASS catalog in our analysis. In addition, we also checked the USNO-B1.0 (Monet et al. 2003) and 2MASS (Cutri et al. 2000) catalogs. We included all the coincidences in Table 1. Since the astrometry of these catalogs is better than $1''$, we used a searching radius of $1''.5$ to search for GCs and planetary nebulae. We found 32 X-ray GCs (25 from the ground and seven from *HST*) of 712 optically identified GCs within S3, while we did not find X-ray–emitting planetary nebulae. The general properties of these X-ray GCs will be discussed in detail in next subsections. There are three (four) X-ray GCs that are also coincident with 2MASS (USNO) sources.

[6] See http://hea-www.cfa.harvard.edu/CHAMP.

TABLE 1

SOURCE LIST

OBJECT	R.A. (J2000.0)	DECL. (J2000.0)	NET COUNTS			L_X[a] ($\times 10^{38}$)	OPTICAL MAGNITUDE	NOTES
			Soft	Medium	Hard			
X1	12 09 38.7	−11 38 51.1	21.7 ± 4.8	11.1 ± 3.4	−0.5 ± 0.2	0.69	$B = 13.8$, $R = 12.5$, $K = 10.9$	Star, QSS-FNOH
X2	12 39 42.3	−11 36 56.1	7.6 ± 2.8	9.4 ± 3.0	6.9 ± 2.6	0.73		
X3	12 39 43.1	−11 36 44.2	2.9 ± 1.8	6.0 ± 2.5	6.7 ± 2.7	0.48		
X4	12 39 44.2	−11 36 00.5	30.3 ± 5.6	21.4 ± 4.6	7.2 ± 2.8	1.76		
X5	12 39 45.2	−11 38 49.5	630.2 ± 25.1	144.9 ± 12.0	28.8 ± 5.3	19.20	$B = 10.3$, $R = 9.7$, $K = 8.4$	Star, QSS-σ
X6	12 39 45.2	−11 36 00.4	36.6 ± 6.1	39.5 ± 6.3	25.0 ± 5.1	2.48	$B = 20.0$, $R = 18.5$, $K = 15.6$	CFHT
X7	12 39 45.6	−11 39 33.1	7.2 ± 2.6	4.9 ± 2.2	10.0 ± 3.1	0.67	$B = 19.9$, $R = 18.9$	Star
X8	12 39 47.2	−11 38 14.6	9.3 ± 3.0	6.3 ± 2.5	4.9 ± 2.3	0.63		
X9	12 39 48.6	−11 37 13.0	87.5 ± 9.3	64.3 ± 8.0	24.2 ± 4.9	5.42		
X10	12 39 50.1	−11 37 41.7	2.8 ± 1.8	3.1 ± 1.8	1.8 ± 1.4	0.24		
X11	12 39 50.5	−11 39 14.5	19.1 ± 4.4	19.5 ± 4.4	13.2 ± 3.6	1.56	$B = 20.2$, $R = 19.0$	Star
X12	12 39 50.9	−11 38 23.3	31.5 ± 5.6	44.4 ± 6.6	27.9 ± 5.3	3.14		CFHT
X13	12 39 51.1	−11 36 47.7	7.6 ± 2.9	9.3 ± 3.0	3.0 ± 1.7	0.59		
X14	12 39 51.1	−11 39 28.4	−0.6 ± 0.2	3.9 ± 2.0	4.9 ± 2.2	0.26		
X15	12 39 51.1	−11 38 33.7	9.3 ± 3.0	5.2 ± 2.2	1.6 ± 1.4	0.40		QSS-3σ_1
X16	12 39 51.9	−11 35 51.9	38.2 ± 6.2	27.5 ± 5.3	8.8 ± 2.9	2.29		CFHT
X17	12 39 53.4	−11 37 09.3	6.1 ± 2.7	8.0 ± 2.8	4.1 ± 2.0	0.57		
X18	12 39 54.0	−11 38 25.1	30.7 ± 5.5	3.0 ± 1.7	0.8 ± 1.0	0.84		SSS-3σ
X19	12 39 54.1	−11 37 13.0	4.5 ± 2.3	4.0 ± 2.0	2.0 ± 1.4	0.33		
X20	12 39 54.6	−11 36 02.5	5.3 ± 2.4	1.0 ± 1.0	2.0 ± 1.5	0.26		CFHT
X21	12 39 54.8	−11 37 30.3	1.5 ± 1.4	6.1 ± 2.5	1.0 ± 1.0	0.21		QSS-FNOH
X22	12 39 54.9	−11 37 08.6	6.0 ± 2.7	7.9 ± 2.9	11.4 ± 3.3	0.78		
X23	12 39 55.1	−11 37 37.0	0.6 ± 1.0	11.3 ± 3.3	20.5 ± 4.5	1.01		
X24	12 39 55.4	−11 38 48.9	17.2 ± 4.1	10.3 ± 3.2	11.8 ± 3.4	1.22		CFHT
X25	12 39 55.5	−11 37 32.1	4.7 ± 2.2	3.9 ± 2.0	1.9 ± 1.4	0.32		
X26	12 39 55.8	−11 34 48.1	19.4 ± 4.5	16.3 ± 4.0	14.9 ± 3.9	1.56	$B = 19.3$, $R = 18.4$	CFHT
X27	12 39 55.8	−11 37 31.4	3.2 ± 2.0	7.2 ± 2.7	3.9 ± 2.0	0.42		
X28	12 39 55.9	−11 37 42.1	1.5 ± 1.4	3.9 ± 2.0	4.9 ± 2.2	0.32		
X29	12 39 56.4	−11 31 54.2	32.6 ± 5.9	40.3 ± 6.3	32.5 ± 5.9	3.07	$B = 19.7$, $R = 18.3$, $K = 15.6$	CFHT
X30	12 39 56.4	−11 37 59.0	17.2 ± 4.1	10.3 ± 3.2	6.0 ± 2.4	1.03		
X31	12 39 56.4	−11 37 24.3	13.9 ± 4.1	11.4 ± 3.4	6.1 ± 2.4	0.99		
X32	12 39 56.5	−11 38 30.5	7.9 ± 2.8	5.1 ± 2.2	10.0 ± 3.1	0.68		
X33	12 39 56.9	−11 36 58.6	3.7 ± 2.0	3.9 ± 2.0	8.1 ± 2.8	0.42		
X34	12 39 57.2	−11 37 03.9	5.9 ± 2.5	1.9 ± 1.4	2.6 ± 1.7	0.32		
X35	12 39 57.2	−11 36 33.6	4.5 ± 2.2	3.1 ± 1.7	0.8 ± 1.0	0.21		CFHT, QSS-NOH
X36	12 39 57.3	−11 36 29.5	5.8 ± 2.5	2.9 ± 1.7	−0.2 ± 0.1	0.19		QSS-NOH
X37	12 39 57.3	−11 37 50.3	6.1 ± 2.5	3.1 ± 1.7	0.0 ± 0.0	0.23		QSS-FNOH
X38	12 39 57.4	−11 37 19.8	51.1 ± 7.1	43.9 ± 6.6	46.4 ± 6.8	4.17		
X39	12 39 57.4	−11 37 52.8	14.4 ± 3.8	12.4 ± 3.5	7.9 ± 2.8	1.07		
X40	12 39 57.4	−11 37 33.6	5.8 ± 2.5	3.0 ± 1.7	5.8 ± 2.5	0.46		
X41	12 39 57.7	−11 38 52.9	11.9 ± 3.5	11.2 ± 3.3	6.9 ± 2.6	0.93		
X42	12 39 57.7	−11 37 14.6	6.9 ± 2.7	3.9 ± 2.0	2.0 ± 1.4	0.40		
X43	12 39 57.9	−11 37 02.3	15.9 ± 4.0	12.2 ± 3.5	3.0 ± 1.7	0.96		
X44	12 39 57.9	−11 37 34.5	9.0 ± 3.0	15.4 ± 3.9	4.8 ± 2.2	0.87		
X45	12 39 58.3	−11 37 17.8	18.3 ± 4.3	10.3 ± 3.2	7.0 ± 2.6	1.06		
X46	12 39 58.5	−11 37 21.2	12.1 ± 3.5	3.9 ± 2.0	6.1 ± 2.4	0.68		
X47	12 39 58.6	−11 37 27.6	7.0 ± 2.7	8.4 ± 2.9	8.0 ± 2.8	0.72		*HST*
X48	12 39 58.7	−11 38 20.6	8.3 ± 2.9	2.1 ± 1.4	−0.1 ± 0.0	0.23		QSS-NOH
X49	12 39 58.7	−11 37 21.4	8.1 ± 2.8	7.2 ± 2.7	3.8 ± 2.0	0.59		
X50	12 39 58.8	−11 37 24.9	27.6 ± 5.3	13.6 ± 3.6	11.3 ± 3.3	1.59		*HST*
X51	12 39 58.8	−11 37 00.2	3.7 ± 2.0	5.0 ± 2.2	2.6 ± 1.7	0.35		
X52	12 39 58.9	−11 36 54.6	9.8 ± 3.2	9.3 ± 3.0	6.9 ± 2.6	0.81		CFHT
X53	12 39 58.9	−11 38 37.9	10.8 ± 3.3	21.4 ± 4.6	8.0 ± 2.8	1.24		CFHT, *HST*
X54	12 39 59.0	−11 35 12.0	24.0 ± 4.9	40.2 ± 6.3	22.08 ± 4.7	2.62		CFHT
X55	12 39 59.1	−11 37 19.4	61.0 ± 7.8	28.9 ± 5.4	10.3 ± 3.2	2.93		*HST*
X56	12 39 59.3	−11 38 28.1	18.3 ± 4.2	20.3 ± 4.5	10.5 ± 3.2	1.51		CFHT, *HST*
X57	12 39 59.3	−11 36 50.0	8.7 ± 3.0	1.8 ± 1.4	−0.3 ± 0.1	0.25		QSS-FNOH
X58	12 39 59.3	−11 38 46.0	1.8 ± 1.4	3.0 ± 1.7	2.9 ± 1.7	0.24		CFHT, *HST*
X59	12 39 59.4	−11 37 26.8	57.4 ± 7.5	39.7 ± 6.3	25.5 ± 5.0	3.78		*HST*
X60	12 39 59.4	−11 37 22.7	709.4 ± 26.6	999.9 ± 31.6	906.5 ± 30.1	80.32		Nucleus
X61	12 39 59.5	−11 37 01.8	22.4 ± 4.7	16.4 ± 4.0	19.5 ± 4.4	1.80		
X62	12 39 59.6	−11 37 29.0	8.7 ± 3.0	6.2 ± 2.5	−0.3 ± 0.1	0.37		QSS-FNOH
X63	12 39 59.7	−11 35 24.8	18.9 ± 4.3	21.2 ± 4.6	19.1 ± 4.3	1.73		CFHT
X64	12 39 59.8	−11 37 16.1	41.3 ± 6.4	10.1 ± 3.2	1.8 ± 1.4	1.33		QSS-HR1
X65	12 39 59.8	−11 34 54.4	10.7 ± 3.3	8.0 ± 2.8	6.2 ± 2.6	0.77		CFHT
X66	12 39 59.8	−11 37 00.2	7.9 ± 2.8	9.2 ± 3.0	4.1 ± 2.0	0.66		
X67	12 39 59.9	−11 36 22.8	36.7 ± 6.1	37.2 ± 6.1	22.4 ± 4.7	2.89		

TABLE 1—*Continued*

Object	R.A. (J2000.0)	Decl. (J2000.0)	Net Counts			L_X[a] ($\times 10^{38}$)	Optical Magnitude	Notes
			Soft	Medium	Hard			
X68	12 39 59.9	−11 37 39.7	2.7 ± 1.7	2.0 ± 1.4	3.0 ± 1.7	0.24		
X69	12 40 00.0	−11 37 26.6	11.2 ± 3.3	11.4 ± 3.3	7.0 ± 2.6	0.91		
X70	12 40 00.0	−11 37 08.5	28.7 ± 5.3	35.0 ± 5.9	19.5 ± 4.4	2.56		
X71	12 40 00.1	−11 35 42.0	13.8 ± 3.7	-0.0 ± 0.0	0.0 ± 0.0	0.14		SSS-HR
X72	12 40 00.2	−11 37 22.6	19.7 ± 4.4	20.8 ± 4.5	13.2 ± 3.6	1.59		
X73	12 40 00.3	−11 37 23.2	54.0 ± 7.3	29.1 ± 5.4	19.4 ± 4.4	3.06		*HST*
X74	12 40 00.4	−11 37 17.6	7.8 ± 2.8	8.1 ± 2.8	1.7 ± 1.4	0.55		
X75	12 40 00.7	−11 37 12.5	8.1 ± 2.8	8.1 ± 2.8	0.7 ± 1.0	0.42		QSS-FNOH
X76	12 40 00.7	−11 35 19.4	31.8 ± 5.6	27.1 ± 5.2	25.1 ± 5.0	2.55		CFHT
X77	12 40 00.7	−11 37 04.3	13.2 ± 3.6	12.3 ± 3.5	15.4 ± 3.9	1.26		
X78	12 40 00.7	−11 37 30.3	3.5 ± 2.0	5.2 ± 2.2	2.9 ± 1.7	0.36		
X79	12 40 00.9	−11 36 53.9	138.3 ± 11.7	111.7 ± 10.5	42.2 ± 6.5	8.87		
X80	12 40 01.0	−11 37 08.3	6.5 ± 2.7	7.1 ± 2.6	5.1 ± 2.2	0.58		CFHT
X81	12 40 01.0	−11 37 00.9	15.0 ± 3.9	2.5 ± 1.7	7.0 ± 2.6	0.69		*HST*
X82	12 40 01.1	−11 37 23.8	391.3 ± 19.7	81.8 ± 9.0	5.9 ± 2.4	11.48		QSS-σ
X83	12 40 01.3	−11 37 02.0	3.9 ± 2.0	6.1 ± 2.4	2.7 ± 1.7	0.39		
X84	12 40 01.3	−11 37 29.8	9.1 ± 3.0	10.4 ± 3.2	16.4 ± 4.0	1.08		
X85	12 40 01.4	−11 40 10.1	2.7 ± 1.7	3.8 ± 2.0	1.9 ± 1.4	0.26		
X86	12 40 02.0	−11 39 40.1	71.5 ± 8.4	0.0 ± 0.0	-0.1 ± 0.1	1.55		SSS-HR
X87	12 40 02.0	−11 37 07.1	6.6 ± 2.6	7.1 ± 2.6	2.8 ± 1.7	0.51		CFHT, *HST*
X88	12 40 02.0	−11 38 46.4	5.1 ± 2.3	1.9 ± 1.4	1.8 ± 1.4	0.28		
X89	12 40 02.2	−11 37 53.8	19.6 ± 4.4	3.9 ± 2.0	10.2 ± 3.2	1.04		
X90	12 40 02.2	−11 37 18.6	14.9 ± 3.9	8.9 ± 3.0	13.1 ± 3.6	1.08		
X91	12 40 02.2	−11 38 01.5	6.7 ± 2.7	3.7 ± 2.0	6.1 ± 2.4	0.51		CFHT, *HST*
X92	12 40 02.3	−11 37 11.7	8.7 ± 3.0	6.2 ± 2.4	1.0 ± 1.0	0.40		SSS-FNOH
X93	12 40 02.4	−11 38 52.3	10.0 ± 3.2	1.9 ± 1.4	3.9 ± 2.0	0.49		
X94	12 40 02.8	−11 37 16.8	11.9 ± 3.5	6.1 ± 2.4	6.0 ± 2.4	0.74		
X95	12 40 03.1	−11 40 04.9	4.8 ± 2.2	6.1 ± 2.4	6.9 ± 2.6	0.55	$B = 18.0$, $R = 16.5$, $K = 14.6$	Star
X96	12 40 03.6	−11 37 39.2	3.7 ± 2.0	7.9 ± 2.9	11.2 ± 3.3	0.70		
X97	12 40 03.7	−11 32 43.0	13.4 ± 3.9	11.0 ± 3.3	13.5 ± 4.0	1.10	$B = 20.7$, $R = 19.6$	Star
X98	12 40 03.7	−11 37 00.9	7.4 ± 2.8	3.0 ± 1.7	0.8 ± 1.0	0.28		QSS-FNOH
X99	12 40 04.5	−11 36 37.5	13.9 ± 3.7	7.4 ± 2.7	6.0 ± 2.5	0.84		
X100	12 40 04.6	−11 37 35.8	2.5 ± 1.7	9.1 ± 3.0	14.3 ± 3.7	0.81		
X101	12 40 04.7	−11 38 28.8	1.2 ± 1.4	4.0 ± 2.0	5.1 ± 2.2	0.33		CFHT, *HST*
X102	12 40 04.8	−11 37 21.0	15.4 ± 4.0	2.7 ± 1.7	1.8 ± 1.4	0.48		QSS-3σ
X103	12 40 05.0	−11 37 32.3	9.0 ± 3.0	8.3 ± 2.8	13.3 ± 3.6	0.91		
X104	12 40 05.3	−11 35 00.0	18.0 ± 4.3	24.2 ± 4.9	17.1 ± 4.3	1.83		CFHT
X105	12 40 05.5	−11 39 40.5	23.0 ± 4.8	1.7 ± 1.4	4.7 ± 2.2	0.82		
X106	12 40 05.7	−11 37 11.2	20.0 ± 4.5	14.3 ± 3.8	8.9 ± 3.0	1.30		*HST*
X107	12 40 06.9	−11 37 22.0	4.4 ± 2.2	3.1 ± 1.7	1.8 ± 1.4	0.30		
X108	12 40 07.0	−11 37 53.2	20.4 ± 4.5	18.7 ± 4.3	21.4 ± 4.6	1.87		CFHT, *HST*
X109	12 40 08.2	−11 34 46.2	3.9 ± 2.3	9.4 ± 3.1	0.8 ± 1.5	0.36		QSS-FNOH
X110	12 40 08.4	−11 37 11.3	10.0 ± 3.2	3.0 ± 1.7	1.8 ± 1.4	0.35		QSS-HR1
X111	12 40 08.9	−11 38 17.2	11.4 ± 3.4	1.9 ± 1.4	0.0 ± 0.0	0.30		SSS-3σ
X112	12 40 09.5	−11 36 46.6	21.5 ± 4.6	12.3 ± 3.5	10.1 ± 3.2	1.32		
X113	12 40 09.7	−11 36 44.7	2.9 ± 1.7	5.1 ± 2.2	2.9 ± 1.7	0.34		CFHT
X114	12 40 10.4	−11 36 38.7	41.4 ± 6.5	33.0 ± 5.7	26.5 ± 5.1	3.01		CFHT
X115	12 40 10.8	−11 32 58.1	22.2 ± 4.9	18.3 ± 4.3	23.4 ± 4.9	1.90	$B = 19.3$, $R = 17.6$, $K = 14.8$	CFHT
X116	12 40 11.4	−11 34 42.9	8.2 ± 3.2	8.2 ± 3.0	12.5 ± 3.7	0.89		
X117	12 40 12.3	−11 40 05.3	6.2 ± 2.7	7.0 ± 2.7	7.5 ± 2.9	0.63		
X118	12 40 12.7	−11 36 30.9	3.9 ± 2.3	3.8 ± 2.0	4.1 ± 2.3	0.37		
X119	12 40 12.9	−11 39 03.0	9.0 ± 3.2	2.8 ± 1.7	3.8 ± 2.0	0.51		
X120	12 40 13.7	−11 37 52.2	13.4 ± 3.8	6.8 ± 2.7	2.6 ± 1.7	0.62		
X121	12 40 15.2	−11 33 07.6	21.3 ± 5.4	50.1 ± 7.2	15.2 ± 4.4	2.63		3 USNO stars
X122	12 40 15.5	−11 37 34.8	6.1 ± 2.6	5.5 ± 2.4	4.8 ± 2.4	0.48		

Notes.—Units of right ascension are hours, minutes, and seconds, and units of declination are degrees, arcminutes, and arcseconds. The columns give the source number, the position (J2000.0), the net counts in the three energy bands (soft: 0.1–1.1 keV; medium: 1.1–2 keV; hard: 2–7 keV), the 0.3–7 keV luminosity. The conversion to luminosities assumes an absorbed power-law spectrum with a photon index of 2 and $N_H = 5 \times 10^{20}$ cm^{-2}. For SSSs and QSSs, we assume a blackbody model with $kT = 125$ eV and $N_H = 5 \times 10^{20}$ cm^{-2}. The *B* and *R* magnitudes are taken from USNO-B1.0 catalog, while the *K* magnitude is from the 2MASS catalog. "CFHT" is for globular clusters from the ground; *HST* is for globular clusters from *HST*; and SSS-HR, SSS-3σ, QSS-HR1, QSS-3σ_1, QSS-σ, QSS-FNOH, and QSS-NOH are for the very soft source category from Stefano & Kong (2003b).

[a] Luminosity in 0.3–7 keV (erg s^{-1}), assuming $d = 8.9$ Mpc, a power-law model with $N_H = 5 \times 10^{20}$ cm^{-2}, and $\alpha = 2$ for all sources except SSSs, for which a blackbody model with $N_H = 5 \times 10^{20}$ cm^{-2} and $kT = 125$eV is applied.

In order to estimate the accidental correlation rate, we shifted all the *Chandra* sources by 4″, 6″, 8″, and 10″ to the north, east, south, west, northeast, northwest, southeast, and southwest, and ran the search for each of the catalogs. The average accidental matching rates for the GC catalog from the ground, the GC catalog from the *HST*, and the planetary nebula catalog are 2.8, 3.9, and 0.7, respectively. We therefore estimate that 22 of the X-ray sources associated with GCs identified from ground-based observations and three from *HST* observations represent genuine matches. The relatively higher accidental matching rate for the *HST* catalog may due to the crowding in the nuclear region of M104. The analyses presented in the remainder of the paper apply only to the GCs observed from the ground.

3.2. *The Nuclear Source*

Pellegrini et al. (2002) used a 1.7 ks ACIS-S observation to study the spectrum of the central source. They found a power-law fit ($\alpha = 1.5^{+0.4}_{-0.3}$), with $N_{\rm H} = 1.7^{+1.1}_{-0.9} \times 10^{21}$ cm^{-2}. The flux was 1.6×10^{-12} ergs cm^{-2} s^{-1}, which translates to $L = 1.5 \times 10^{40}$ ergs s^{-1} at 8.9 Mpc. During the 18.5 ks observation, photons from the nuclear source were piled up ($\sim$20%). By using a pileup model developed for *Chandra* data (Davis 2001), we found a power-law fit ($\alpha = 2.18^{+0.35}_{-0.35}$), with $N_{\rm H} = 2.9^{+0.57}_{-0.57} \times 10^{21}$ cm^{-2}; assuming a distance of 8.9 Mpc, $L = 2.0 \times 10^{40}$ ergs s^{-1}. The properties of the nuclear source in the second, longer observation we report on are roughly consistent with those described by Pellegrini et al. (2002). They are also consistent with an analysis including *XMM-Newton* data (see Pellegrini et al. 2003a for a more complete discussion of the nuclear source).

3.3. *Spectral Fits of Bright Sources*

We extracted the energy spectra for the six brightest sources (>100 counts) and fitted them to simple one-component spectral models including absorbed power-law, blackbody, disk blackbody, and Raymond-Smith models. The background spectrum was extracted from source-free regions around each source. We also corrected for the degradation of soft energy (<1 keV) sensitivity.[7] Table 2 lists the spectral fits of these sources. Except for X5, which is a bright foreground star ($B = 10.3$), all sources can be fitted with a single-component model. The spectral shapes of X9, X38, X59, and X79 are consistent with an absorbed power-law model with photon index between 1.1 and 2. These spectra are similar to typical X-ray binaries seen in our own Galaxy and in other galaxies. Figure 3 shows the energy spectrum of the brightest point source (excluding the nucleus), X79. X82 is soft compared with the other sources; the energy spectrum is a blackbody model with thermal temperature of 0.18 keV (see Table 2 and Fig. 3), and it is also classified as a QSS (see above). The foreground star, X5, is soft and requires a Raymond-Smith plus power-law model (see Table 2). The values of $N_{\rm H}$ are generally low and are roughly consistent with the Galactic value along the direction to M104 ($N_{\rm H} = 3.7 \times 10^{20}$ cm^{-2}). The 0.3–7 keV luminosities of these sources (excluding X5) range from 4×10^{38} to 1×10^{39} ergs s^{-1} at a distance of 8.9 Mpc.

[7] See http://asc.harvard.edu/ciao/threads/apply_acisabs.

TABLE 2
SPECTRAL FITS TO THE BRIGHTEST X-RAY SOURCES IN M104

Object	$N_{\rm H}$ (10^{20}cm^{-2})	α	$kT/kT_{\rm RS}$ (keV)	χ^2_ν/dof	$L_{\rm X}$[a]
X5.....	11.1^{+10}_{-6}	$2.86^{+0.68}_{-0.57}$	$0.36^{+0.05}_{-0.06}$[b]	1.64/25	...[c]
X9.....	$6.1^{+7.6}_{-6.0}$	$2.0^{+0.45}_{-0.47}$		0.8/5	6.34
X38.....	$2.1^{+8.4}_{-2.1}$	$1.13^{+0.39}_{-0.32}$		0.85/10	5.89
X59.....	$5.1^{+10}_{-5.1}$	$1.78^{+0.35}_{-0.53}$		0.88/8	4.27
X79.....	$10.8^{+6.6}_{-6.7}$	$2.04^{+0.43}_{-0.28}$		0.83/10	12
X82.....	$0.9^{+4.6}_{-0.9}$		$0.18^{+0.02}_{-0.02}$	0.71/15	8.9

NOTE.—All quoted uncertainties are at the 90% confidence level.
[a] Unabsorber luminosity ($\times 10^{38}$ ergs s^{-1}) in 0.3–7 keV, assuming a distance of 8.9 Mpc.
[b] Raymond-Smith temperature.
[c] X5 is a foreground star. Assuming a distance of 4 kpc, the 0.3–7 keV luminosity is 5.36×10^{32} ergs s^{-1}.

3.4. *X-Ray Luminosity Function*

We constructed the X-ray point-source luminosity function (LF). The count rates for all detected sources were converted into unabsorbed 0.3–7 keV luminosities by assuming an appropriate spectral model. For the brightest six sources (Table 2), we used the luminosities derived from the best-fitting spectral model. In addition, we excluded all the foreground stars (see Table 1). In Figure 4 we plot the cumulative luminosity functions for all sources (excluding GCs) and X-ray GCs. Because we have eliminated foreground stars and expect that fewer than 10 of the X-ray sources are background AGNs (see § 2), these luminosity functions should primarily describe the properties of sources associated with M104.

To estimate the completeness limit, we used a method similar to that of Kong et al. (2002a 2003a). We computed histograms of the number of detected sources against the signal-to-noise ratio to examine the completeness limit; the histograms peak at S/N $\sim$ 3.2, corresponding to 4.5×10^{37}ergs s^{-1}, and fall off below this. Hence, the luminosity function is complete down to 4.5×10^{37} ergs s^{-1}. It is clear that the two luminosity functions are not well described by a simple power law. Instead, they both have a break or cutoff at near 1–3 $\times 10^{38}$ ergs s^{-1}. In addition, power-law models may not be good descriptions above and/or below the breaks. We nevertheless use a broken power-law model for each LF, to facilitate comparison with other treatments (see, e.g., Sarazin et al. 2001). The function is defined as

$$\frac{dN}{dL_{38}} = A\left(\frac{L_{38}}{L_b}\right)^{-\Gamma}, \qquad (1)$$

where $\Gamma = \Gamma_1$ for $L_{38} \le L_b$, $\Gamma = \Gamma_2$ for $L_{38} > L_b$, and L_{38} is the 0.3–7 keV X-ray luminosity in units of 10^{38} ergs s^{-1}. We used a maximum-likelihood method (e.g., Crawford, Jauncey, & Murdoch 1970) to fit the differential luminosity functions of point sources (excluding nucleus, GCs, and stars) and X-ray GCs with a broken power law. For X-ray point sources, the luminosity break is at $9^{+2.0}_{-1.6} \times 10^{37}$ ergs s^{-1}, while $\Gamma_1 = 0.74^{+0.84}_{-1.04}$, $\Gamma_2 = 2.58^{+0.35}_{-0.28}$, and $A = 45.85$. For X-ray GCs, the shape is nearly a cutoff power law with a luminosity break at $3.1^{+0.26}_{-0.24} \times 10^{38}$ ergs s^{-1}, $\Gamma_1 = 0.67^{+0.36}_{-0.39}$, $\Gamma_2 = 8.6^{+4.6}_{-3.2}$, and $A = 5.90$.

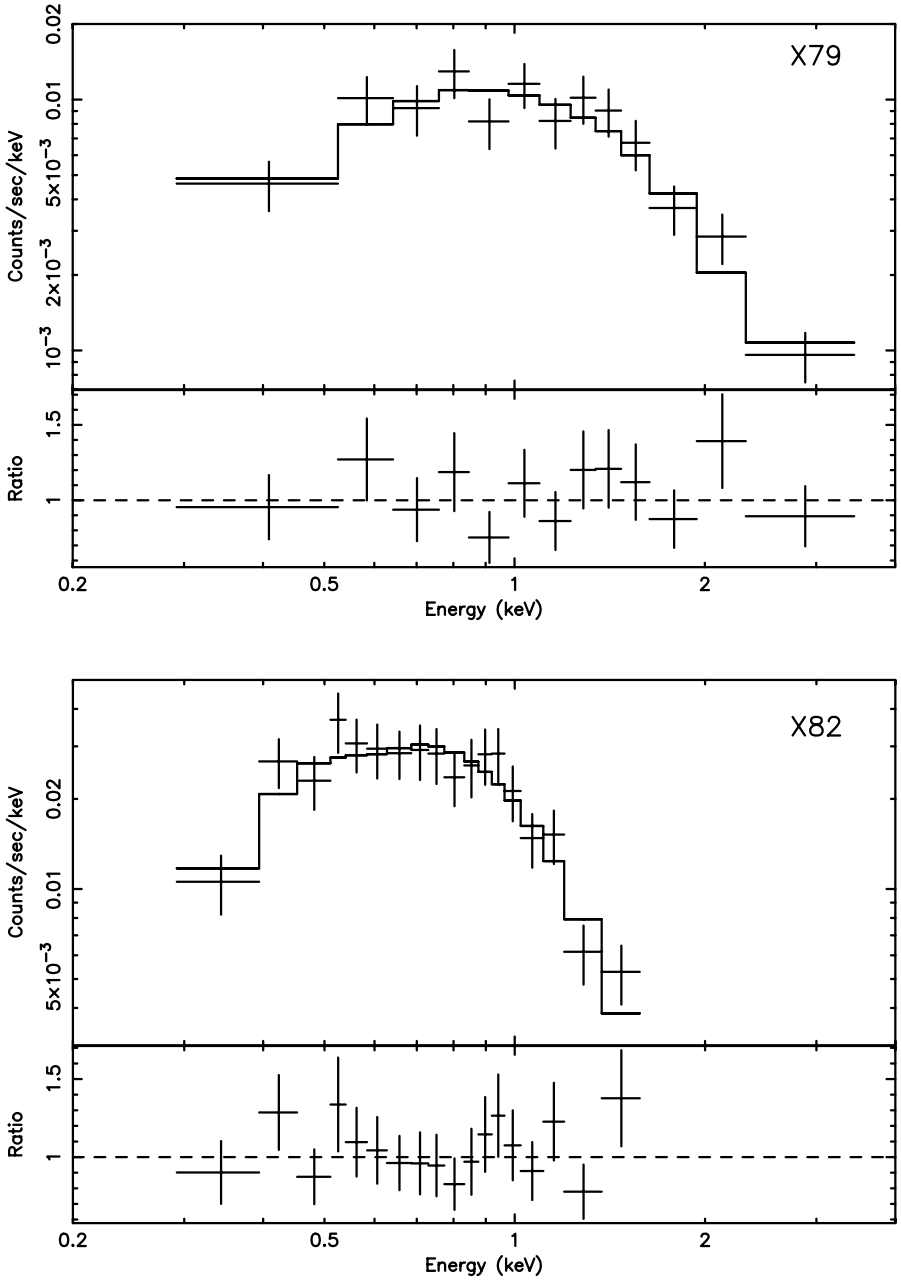


FIG. 3.—Spectral fit to X79 (power-law model with $N_{\rm H} = 1.1 \times 10^{21}$ cm^{-2} and $\alpha = 2.04$) and X82 (blackbody model with $N_{\rm H} = 0.85 \times 10^{20}$ cm^{-2} and $kT = 0.18$ keV).

3.5. *Color-Color Diagram*

Many of the detected sources provide fewer than 100 counts, which makes it difficult to derive spectral parameters. However, hardness ratios can give a crude indication of the X-ray spectra. We therefore computed the hardness ratios for each detected sources. We used three energy bands: S (0.1–1.1 keV), M (1.1–2 keV), and H (2–7 keV). We define HR1 = (M − S)/(M + S) and HR2 = (H − S)/ (H + S), to plot the color-color diagram as shown in Figure 5. We calculated the 1 σ uncertainties of the hardness ratios by using a maximum likelihood method as used in the *Einstein* catalog (Harris et al. 1993).

4. VERY SOFT X-RAY SOURCES

SSSs have luminosities in the range 10^{35}–10^{39} ergs s^{-1} and kT in the range 10–100 eV. SSSs in our Galaxy are largely obscured from us by the interstellar medium (ISM); it is therefore possible that this poorly understood class is in fact the dominant class of high-luminosity X-ray binaries. Di Stefano & Rappaport (1994) estimated, on the basis of *ROSAT* observations and simple models of galaxy absorption, that the Milky Way and M31 may each harbor on the order of 1000 SSSs with $L_{\rm X} > 10^{37}$ ergs s^{-1} and $kT > 30$ eV. SSSs may significantly influence their galactic environments, both as efficient ionizers of the interstellar medium and possibly also as progenitors of Type Ia supernovae.

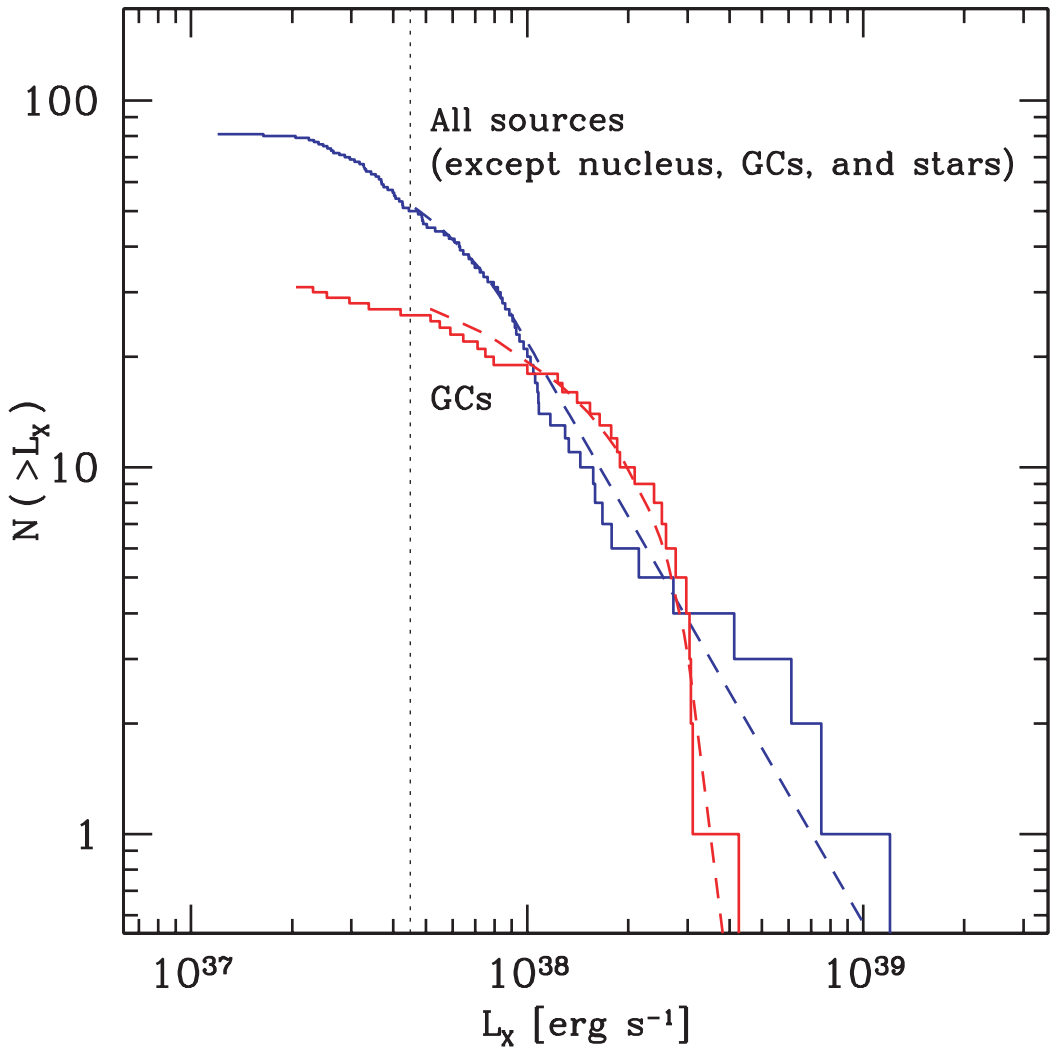


FIG. 4.—Cumulative luminosity functions and their best-fit model of X-ray point sources (*blue*; excluding the nucleus and stars) and GCs (*red*) in M104. The vertical dotted line represents the completeness limit (4.5×10^{37}ergs s^{-1}) of our data.

Some SSSs are simply hot white dwarfs (WDs; e.g., post-novae), pre–white dwarfs (in planetary nebulae), or symbiotics. However, the physical nature of a majority of the SSSs with optical identifications is not yet understood. These more mysterious sources include CAL 83 and CAL 87 (Long, Helfand, & Grabelsky 1981), as well as seven sources discovered with *ROSAT* (see Greiner 2000 for details). Black hole (BH; Cowley et al. 1990) and accreting neutron star models (Greiner, Hasinger, & Kahabka 1991; Kylafis & Xilouris 1993) may apply to some SSSs. Binary models that predict quasi-steady nuclear burning of accreted matter on the surface of a WD are considered promising (see, e.g., van den Heuvel et al. 1992.) Nevertheless, studies of external galaxies find some SSSs that may be BHs with $M \sim 100\ M_{\odot}$ (Di Stefano & Kong 2003a, 2003b).

The primary goals that we hope to achieve by conducting SSS searches in external galaxies are to gain an understanding of the physical nature or natures of the sources and to eventually improve estimates of the sizes of galaxy populations of SSSs. We gain important insight by being able to study the locations of SSSs relative to large-scale galaxy structures (bulge, disk, and halo), to study the stellar populations within which SSSs are embedded, to establish luminosity distributions and spectral characteristics for large samples of sources, and to estimate total populations as a function of galaxy properties. A challenge that must be met by any such program of study is that the small number of counts that we receive from X-ray sources in external galaxies makes spectral classification difficult. Selection is further complicated by the fact that we have only a small sample of local (Galaxy and Magellanic Cloud) sources, whose properties may not span the full range of SSS properties. Di Stefano & Kong (2003a, 2003b, 2003c) have developed an algorithm to select SSSs in external galaxies. The application of our algorithm to real data has led to some intriguing results.

First, we have found it difficult to use a simple algorithmic approach to enforce a strict temperature cut-off. Our algorithm should therefore identify "classical" SSSs like those in the Magellanic Clouds, as well as slightly harder sources with $kT < 300$ eV. We call the latter quasi-soft sources (QSSs). The selection and classification scheme is discussed

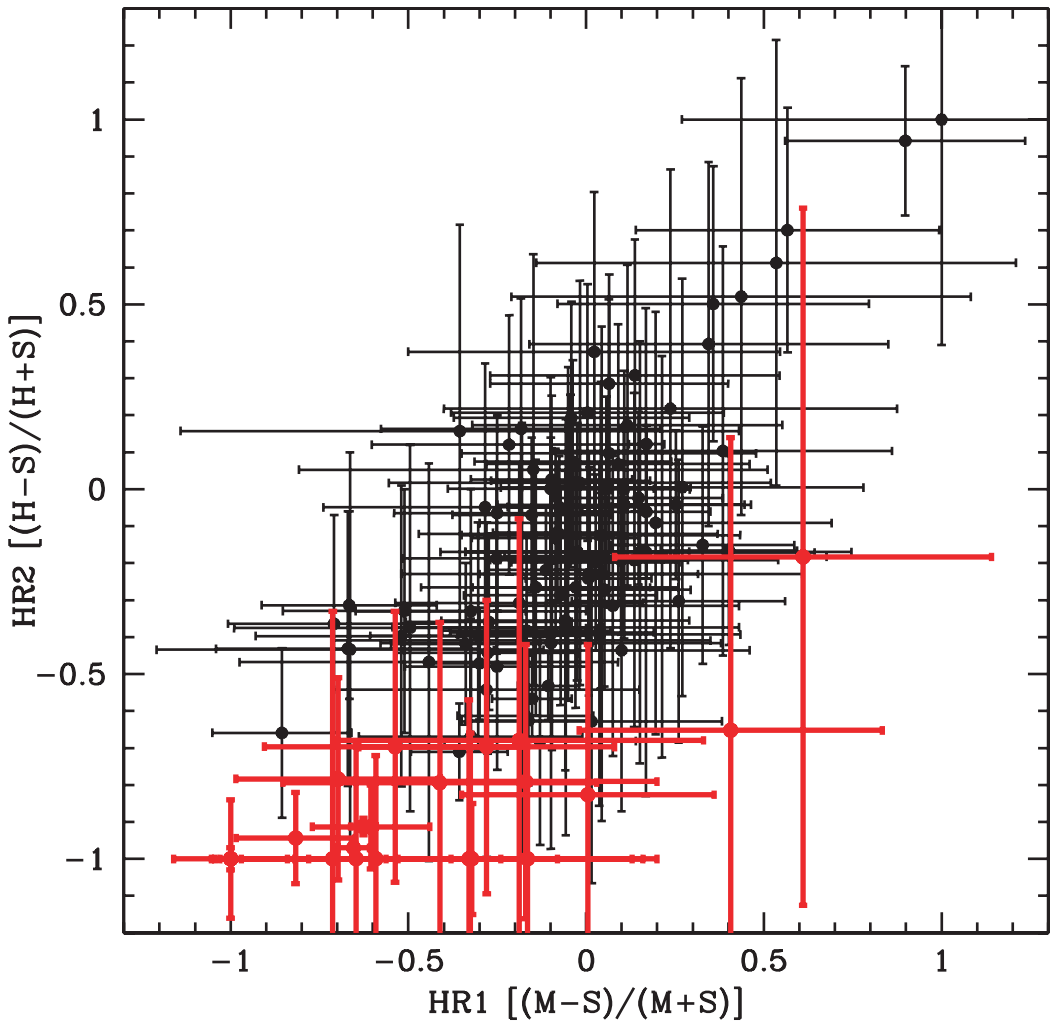


FIG. 5.—Color-color diagram of all X-ray point sources in M104. Very soft sources are marked with red lines. Some of these are SSSs, and some are QSSs (see text).

in detail in Di Stefano & Kong (2003a, 2003b, 2003c). Here we simply note that sources marked either "HR" or "3σ" are likely to have spectra similar to those of the SSSs studied so far in the Magellanic Clouds and Milky Way. We refer to them as classical SSSs. Sources labeled "FNOH" or "NOH," may be highly absorbed or have somewhat higher temperatures, but kT is still smaller than approximately 250 eV. Sources marked "σ", "$3\sigma_1$," or "HR1" could have temperatures in the same general range as "FNOH" or "NOH" sources; our studies of SSSs in other galaxies show, however, that some have spectra with dominant components as soft as the "HR" or "3σ" sources, but also include a small (<10%) harder component. We refer to the sources in these other categories as QSSs. Note, however, that if we were able to collect enough photons for spectral fits, some QSSs may have classical SSS spectral profiles, while some SSSs may have somewhat harder spectra than is indicated by their simple broadband classification.

In M101, very soft sources appear to be primarily associated with the spiral arms; this, combined with the locations of several SSSs and QSSs near H II regions and/or OB associations, suggests that some very soft sources are relatively young systems. In M31 (but not in M101), there is a bright bulge population of SSSs and QSSs, including one source within a few parsecs of the nucleus. This suggests that some very soft sources within ~1 kpc of a massive BH may be the stripped cores of giants that have been tidally disrupted by the BH (Di Stefano et al. 2001). Several galaxies house ultraluminous SSSs (with $L_X > 10^{39}$ ergs s^{-1}) that may be accreting intermediate-mass (~50–10^4 $M_\odot$) BHs. These candidate BH spectra typically include some photons with energies above 1.1 keV, even though 90% of the flux is carried by photons of lower energy. Elliptical galaxies (e.g., NGC 4697; Sarazin et al. 2001) also contain SSSs and QSSs, including very soft sources in GCs (NGC 4472; Friedman et al. 2002).

We applied our soft-source selection algorithm to the X-ray sources that we identified in M104 (see Table 1), and roughly one in six X-ray sources in M104 satisfies the conditions designed to identify SSSs and QSSs. Below we discuss the luminosity distribution of the very soft sources detected in M104 and an intermediate-mass BH model for the brightest QSS in M104. The unique feature of M104 is that one can identify and study the disk, bulge, and halo populations of very soft sources; these are discussed in §§ 4.3–4.5.

4.1. *Luminosity Distribution of Very Soft Sources*

The dimmest GC X-ray source (2×10^{37} ergs s^{-1}) and the six dimmest non-GC X-ray sources (starting at 1.4×10^{37} ergs s^{-1}) are all very soft sources. In fact, of 82 non-GC X-ray sources, 15 of the 30 least luminous ones are very soft sources; these represent three-quarters of the nonstellar very soft sources. The dim very soft sources each provide 8–20 photons.

We expect low-count-rate very soft sources for two reasons. First, the intrinsic very soft source luminosity function of some SSSs and QSSs is dominated by low-luminosity sources. The luminosity of nuclear-burning WDs, e.g., increases with increasing WD mass, with L nearing the Eddington limit and kT perhaps in excess of 100 eV, as the WD mass approaches the Chandrasekhar mass. Since, however, WDs with lower mass are more common, most nuclear-burning WDs should be less luminous. In addition, Greiner et al. (1999) and Greiner & Di Stefano (1999), established that there is a lower luminosity extension of the class of SSSs, which presumably corresponds to nuclear-burning WDs of even lower mass than those in the systems first observed with *Einstein* and *ROSAT*. The number of systems in our Galaxy capable of appearing as such lower luminosity SSSs may be approximately 10,000. Second, the effects of absorption on the radiation emitted by very soft sources is severe. The 18.5 ks observation of M104 was likely to only detect those SSSs and QSSs on the outer edge of the side of the disk and bulge nearest to us.

In spite of the numerical dominance of the low count rate very soft sources, SSSs are also well represented among the most luminous X-ray sources. Three of the 12 brightest M104 X-ray sources are SSSs—these include one of the brightest nonnuclear sources, X82.

4.2. *Intermediate-Mass BH Model for the Brightest SSS*

X82 is an QSS located in the bulge of M104. The spectral fit is shown in Figure 3. The spectral parameters (Table 2) are $kT = 180$ eV, $L_{\rm X} = 8.9 \times 10^{38}$ ergs s^{-1}, and $N_{\rm H} = 0.9^{+4.6}_{-0.9} \times 10^{20}$ cm^{-2}. This fit cannot provide a unique physical interpretation. It is, however, consistent with the model of an intermediate-mass BH. Consider a geometrically thin but optically thick accretion disk, and identify the inner edge of the disk with the last stable orbit around an accreting BH. If we assume an emission efficiency of 10% (i.e., that 10% of the rest energy of matter falling into the BH is radiated by the disk), then

$$M_{\rm BH} = 10^3\ M_\odot \left(\frac{49\ {\rm eV}}{kT}\right)^2 \left(\frac{L}{2.9 \times 10^{37}\ {\rm ergs\ s^{-1}}}\right)^{1/2}. \quad (2)$$

For the spectral parameters derived in the fit, we find that the mass of the BH would be $\sim$400 $M_\odot$. This is likely to be a lower limit, since spectral hardening effects, orientation effects, and spin would all tend to increase the derived value of the mass.

4.3. *Very Soft Sources in the Nuclear Region of M104*

There is a clear overdensity of very soft sources in the region near the nucleus: four very soft sources are located within $\sim$1 kpc of the galaxy center, and an additional three SSSs are located within 2 kpc of the center. Since this places one-third of the very soft sources in roughly 2% of the area containing very soft sources, it is clear that this overdensity implies that some of these seven very soft sources are physically close to the nucleus. In M31 there are very soft sources within several parsecs of the nucleus, and members of the bulge population are among the most luminous SSSs and QSSs in the galaxy. While it is possible that all or most of these sources are simply descended from the stellar populations that inhabit the bulge, the presence of SSSs and QSSs very close to the nucleus of M31 has suggested that some bulge very soft sources may be the stripped cores of stars that have been tidally disrupted by a central BH. If this is so, then galaxies with massive BHs, such as M104, with an estimated BH mass of 10^9 $M_\odot$, should have a central overdensity of very soft source.

4.4. *Very Soft Sources in the Disk of M104*

Given the small numbers of SSSs and QSSs observed near the disk, coupled with the large size of the bulge, it is difficult to definitely identify very soft sources as members of a disk population. Indeed, very soft sources in or near the disk are the ones most likely to be subject to absorption. The locations of X21 (nine counts), X37 (nine counts), X110 (15 counts), and possibly X102 (20 counts) are consistent with membership in the disk. Indeed the distributions of counts in S, M, and H for at least three of these sources (X37, X110, and X102) would be consistent with what is expected for a highly luminous, highly absorbed source with $kT \sim 100$ eV.

4.5. *Very Soft Sources in the Halo of M104*

One of the most exciting aspects of the M104 observations is that for the first time, we can unambiguously identify very soft sources located out of the disk of a spiral galaxy. (See also Di Stefano et al. 2003a, on M31.) The two softest sources, X71 and X86, are each located more than 4 kpc away (north and south, respectively) from the galactic disk. Neither is associated with a known GC or with any other optical counterpart on the DSS image. From neither source have we received photons with energies greater than 1.1 keV. The estimated luminosity of X71 is 1.4×10^{37} ergs s^{-1}, while for X86 we derive 1.6×10^{38} ergs s^{-1}. (These estimates use a model with $kT = 125$ eV and $N_{\rm H} = 5 \times 10^{20}$ cm^{-2}.) If these are not foreground or background objects, they may be good candidates for the nuclear-burning white dwarf model. Since such systems should not have experienced prior supernova explosions, they are unlikely to have been ejected from the disk. They are therefore likely descendants of an old halo stellar population. A particularly interesting QSS is X35, which is in a GC. (From its position alone, it could be part of the bulge or halo.) This source does emit photons between 1.1 and 2 keV. The source properties are consistent with those of a BH accretor with mass equal to $\sim$50 $M_\odot$ (see eq. [2]). This is similar to the situation we have found in NGC 4472, where at least five very soft sources are associated with GCs, several amenable to a similar interpretation (Friedman et al. 2002; Di Stefano et al. 2003b).

4.6. *Relationship between Very Soft Sources and Planetary Nebulae*

In the region covered by the S3 CCD, 249 planetary nebulae (PNe) have been identified. None of these is near enough to an X-ray source to suggest a physical association. The fact that no PNe are associated with very soft sources is

interesting. Because very soft sources emit highly ionizing radiation, they can ionize large ($\sim$10 pc) regions of the surrounding ISM, creating impressive ionization nebulae (Rappaport et al. 1994; Remillard, Rappaport, & Macri 1995). Furthermore, some very soft sources are expected to be PNe. Indeed, one SSS in the Magellanic Clouds is a PN (Wang 1991). SSSs and QSSs that are binary systems can ionize the interstellar medium in which they are embedded. Those located in regions in which the ISM has a density greater than $\sim$1 atom cm^{-3} can create circumstellar ionization nebulae with very large (radiation-limited) radii (>10 pc; Rappaport et al. 1994). CAL 83, located in the Magellanic Clouds, is embedded in such a nebula (Remillard et al. 1995). With two of 12 Magellanic Cloud SSSs associated with nebulae, and with $\sim$1000 SSSs predicted in galaxies such as M31, we might expect there to be dozens of such associations. The nebulae could be and would likely be included among PNe lists by surveys using the methodology that Ford et al (1996) used to identify PNe in M104 (Di Stefano, Paerels, & Rappaport 1995). Because only a small fraction of all SSSs can be detected, the number of observable PNe/very soft sources may be only 1%–10% of the number of physical associations. We might expect some matches for M104 (Di Stefano et al. 1995). The lack of any such associations may mean that very soft sources are typically in regions that have a very low-density ISM, or that they typically are active for periods smaller than the recombination time (typically $\sim 10^5$ yr), with long gaps between "on" times (Chiang & Rappaport 1996). But since it is not clear that either of these possible explanations should or does apply, it would be helpful to push the completeness limit for very soft sources to lower X-ray luminosities. If the link between nebulae and very soft sources can be established and if, as predicted (Rappaport et al. 1994; Di Stefano et al 1995), very soft source nebulae can be distinguished from PNe by a careful examination of the full spectrum of excitation lines, then the nebulae may provide an independent guide to the size of very soft source populations in external galaxies.

4.7. *Relationship between Very Soft Sources and Supernova Remnants*

The very soft X-ray emission of SSSs and QSSs can resemble X-ray supernova remnants (SNRs). There are, e.g., two SSSs in M31 that were subsequently identified as SNRs based on extended morphology and multiwavelength observations (Di Stefano & Kong 2003a; Kong et al. 2002b, 2003b). These two SNRs appear to be constant X-ray sources, which is consistent with the expected picture of SNRs. On the other hand, most of the very soft sources in external galaxies are X-ray variables (Di Stefano & Kong 2003a; Kong & Di Stefano 2003). Although it is extremely difficult to study the morphology of X-ray sources at the distance of M104, and high-resolution narrowband optical imaging is not available, we can study the time variability of these X-ray sources to investigate their nature.

5. GC X-RAY SOURCES

5.1. *Significance*

Galactic GCs house approximately 100 times as many X-ray sources as would be expected on the basis of their mass, suggesting that the efficiency of forming X-ray sources must be higher in GCs than elsewhere in the Galaxy (Clark 1975). A possible explanation, put forward early in the study of these sources, is that the high stellar density near the cluster centers (as large as 10^6 stars pc^{-3}) can lead to stellar interactions in which close binaries containing WDs and neutron stars (NSs) can be formed. Eventually, either because dissipative processes (gravitational radiation or magnetic braking) bring the companion stars close enough for contact to be established, or else because the noncompact star begins to evolve and comes to fill its Roche lobe, mass transfer begins. Because the mass of the donor is generally limited by the turnoff mass ($\sim$0.8 $M_\odot$), these systems are classified as low-mass X-ray binaries (LMXBs). Detailed calculations of interactions that form close binaries, follow the subsequent binary evolution, and include interactions that may occur after the close binaries are formed show that a combination of two-, three-, and four-body interactions can create X-ray binaries and lead them to be ejected to the outer parts of the cluster. It is considered likely that close binaries can be ejected from GCs, since the amount of energy needed to escape the cluster is a small fraction of the binary's binding energy.

LMXBs can also be formed in the field, because a subset of all primordial binaries is expected to evolve into close binaries in which a neutron star accretes mass from a low-mass companion. Before a primordial binary can turn on as an X-ray binary, several processes must occur. The formation of the neutron star occurs quickly, but the spiral-in of the companion or the companion's stellar evolution can require several billion years. Because there are many uncertainties in the evolution of these systems and in the evolution of galaxy populations, it is difficult to accurately predict the numbers of LMXBs expected per unit mass due to a single burst of early star formation. It has been suggested, however, that such systems may not be able to produce enough LMXBs to explain the numbers currently being discovered in early-type galaxies. (It is important to note that some of the early-type galaxies that we observe may have had more complicated histories, with multiple epochs of star formation.) If not, then it is possible that the LMXBs we do see are systems that have been ejected from GCs. Indeed, testing this hypothesis has been an important motivator for study of the GC/LMXB connection. Meaningful tests must include the study of a set of galaxies, including spirals, to search for evidence of differences in the LMXB/GC connection as exhibited by early- and late-type galaxies. Until now, M31 was the only spiral with bright enough GC X-ray sources to enable a comparison with the X-ray source population in distant ellipticals. M104 is therefore an important addition to the set of galaxies in which we can hope to study the LMXB/GC connection.

Below we focus our discussion on the GCs studied from the ground which have been matched with X-ray sources.

5.2. *Optical Selection of Globular Clusters*

Deep wide-field images of M104 have been acquired with the CFHT12K mosaic camera at the Canada-France-Hawaii Telescope (CFHT) in order to investigate the optical characteristics of its large globular cluster system (GCS). A complete description of this photometry will appear in VanDalfsen et al. (2003). In summary, the GCS exhibits a classically bimodal color distribution, with "blue" (metal-poor) and "red" (metal-rich) subpopulations centered at

corresponding metallicities of [Fe/H](blue) = −1.1 and [Fe/H](red) = −0.1. The redder population—which makes up about one-third of the total GCS—is more centrally concentrated in spatial distribution than the bluer clusters.

In terms of the luminosity limit of our optical survey, our data reach nearly 2 mag fainter than the normal "turnover" point (peak frequency) of the globular cluster luminosity function (GCLF). The GCLF turnover is at an absolute visual magnitude $M_V = -7.6$, corresponding to a cluster mass of $M \simeq 1.3 \times 10^5\ M_\odot$, with little difference between the blue and red clusters. In short, the characteristics of the GCS strongly resemble those of the "standard" Milky Way GCS, simply scaled up to a total cluster population 7 or 8 times larger.

5.3. *Properties of Matches*

The procedure we have used to match GCs with X-ray sources is described in § 3. We find 25 matches.

1. Of these, 19 objects lie in the metal-rich side, while six objects lie in the metal-poor side, giving a 1 : 3 ratio of blue : red objects. However, a random sampling of globular clusters would globally yield a 2 : 1 ratio of blue : red objects, approaching a 1 : 1 ratio of blue : red very close to M104. There also seems to be a lack of matches that are both bright and blue. We therefore find a result similar to others reported for NGC 4472 (Maccarone, Kundu, & Zepf 2003; Kundu, Maccarone, & Zepf 2002) and for the Milky Way and M31 (Bellazzini et al. 1995; but see also Di Stefano et al. 2002), i.e., that X-ray sources are preferentially found in GCs that are metal-rich. (See Fig. 6)

2. Figure 7 illustrates the relationship between X-ray luminosity and the magnitude and color of the clusters housing X-ray sources. Two new features are worthy of note. First, the most luminous clusters in both B and R that have X-ray sources house only highly luminous X-ray sources ($L_X > 10^{38}$ ergs s^{-1}). (In contrast, some clusters that

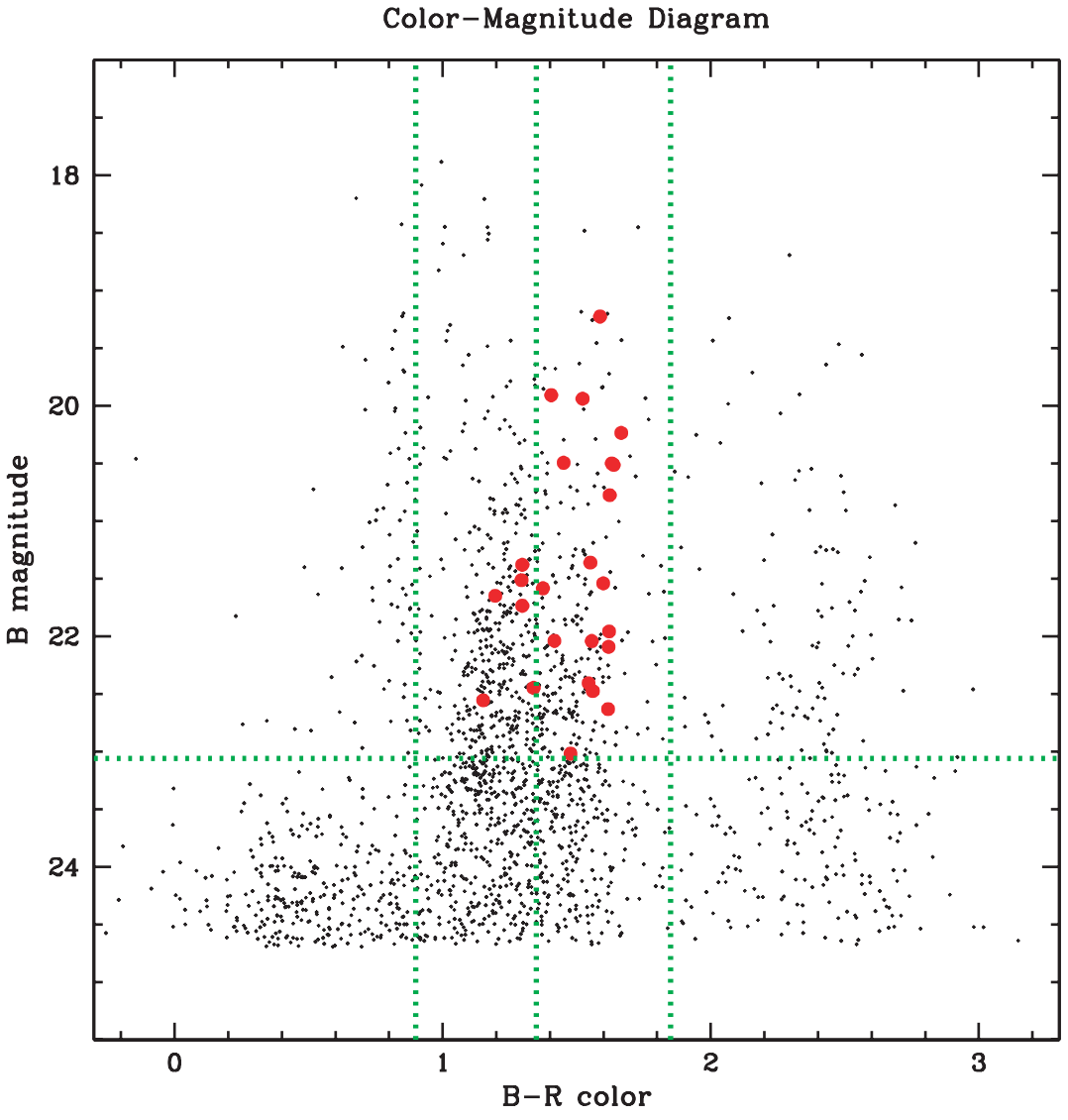


FIG. 6.—Color-magnitude diagram of all of the objects detected by the CFHT survey located within about 9′ (radius) of M104. This diagram shows globular clusters and noncluster contaminants. The 25 X-ray matches are shown by large circles. The horizontal line (at $B = 23.06$) is the location of the GCLF turnover magnitude. The left and right vertical lines are the color limits used to identify GCs; they extend roughly 3 σ out from the GCS population, covering $(B-R)$ from 0.90–1.85. The middle vertical line represents the division between the blue (metal-poor) and red (metal-rich) populations, at $(B-R) = 1.35$.

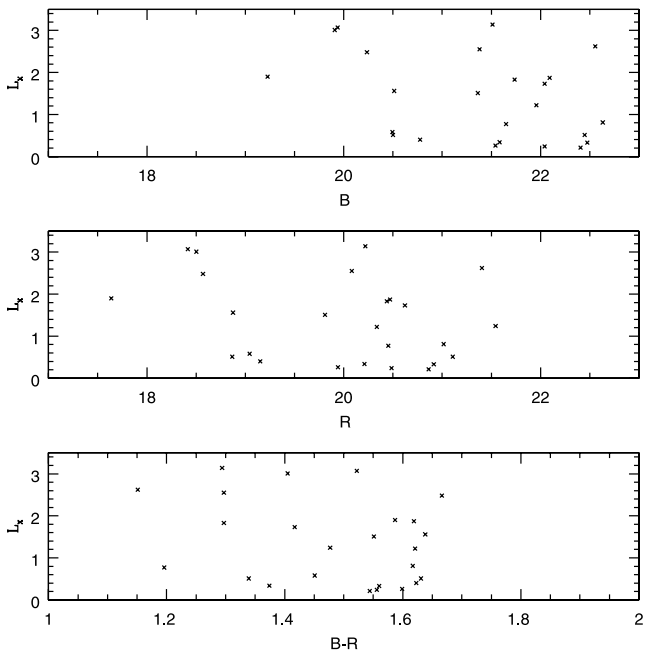


FIG. 7.—X-ray luminosity, in units of 10^{38} ergs s^{-1}, is plotted vs. B magnitude (*top*), R magnitude (*middle*), and $B-R$ (*bottom*). Each point represents a single association between an X-ray source with X-ray luminosity L_X and an CFHT-detected GC with optical properties described by B, R, or $B-R$.

are not as optically bright house X-ray sources of lower luminosity, while others harbor bright X-ray sources.) Second, for X-ray sources with L greater than approximately 2.0×10^{38} ergs s^{-1}, there is no obvious preference for GCs that are optically redder.

3. None of the GC X-ray sources have luminosities greater than the Eddington limit for a 1.4 $M_\odot$ object. Given the small numbers of sources at the highest luminosities and the fact that transient behavior is common in LMXBs, this situation could be different in future observations.

4. Although the four brightest X-ray sources are not in GCs, GC X-ray sources dominate [and in fact represent $\sim$68% of all sources in the luminosity range $\sim(1.2\text{–}4) \times 10^{38}$ ergs s^{-1}]. That is, bright X-ray sources are most likely to be found in GCs.

5. As described in § 4, one GC houses a soft X-ray source, X35, that could be an accreting BH with a mass of approximately 50 $M_\odot$.

5.4. *Interpretation*

The four brightest X-ray sources in M104 are super-Eddington for a 1.4 $M_\odot$ object; these four sources are not located in GCs. Nevertheless, the majority (68%) of all X-ray sources with 1.2×10^{38} ergs $s^{-1} < L_X < 4 \times 10^{38}$ ergs s^{-1} are in GCs. Were M104 located far enough from us, the majority of sources that we observe could be in GCs. M104 provides a second example of a spiral galaxy (M31 being the first) in which the majority of the brightest sources are located in GCs.

Our data are consistent with previous findings that X-ray sources are found preferentially in red, hence metal-rich, GCs. An interesting twist, however, is that this preference for red GCs is not expressed by the brightest X-ray sources. The least luminous GC X-ray sources in our sample do, on the other hand, have a strong preference for red GCs.

5.5. *Models*

Below we discuss possible explanations for the trends apparent in the data.

5.5.1. *Thermal Timescale Mass Transfer*

The properties of the luminous X-ray source in the M31 GC Bo 375 suggested a new type of GC binary evolution model (Di Stefano et al. 2002). In this model, mass transfer is driven by the thermal timescale readjustment of the donor star to changes in the size of its Roche lobe. Mass transfer typically starts when the donor star has a mass that is within a few tenths of a solar mass of the NS and is also slightly evolved (with core mass $\sim$0.1–0.25). Because the donor is more massive than 0.8 $M_\odot$, this type of system can be found only in younger GCs and/or in GCs with such a high rate of stellar interactions that at least one NS can be expected to have a blue straggler donor. One proposed test of this model was to check whether bright X-ray sources in the GCSs of external galaxies are likely to be found in young and/or optically bright GCs. This is exactly what we find in the Sombrero. Thus, although the model is not proved, M104 does provide some indirect evidence in its favor.

5.5.2. *Multiplicity and Ejection*

The correlation between optically bright GCs and high-luminosity X-ray sources is intriguing. It is likely that the large optical flux is indicative of a massive GC, in which stellar interactions producing X-ray binaries may be more frequent. More frequent interactions may lead to more X-ray sources per cluster. There are two possible checks for multiplicity.

Time variability.—GCs that do not house multiples, or that include only a small number ($\sim$2–3) of individual components, could exhibit significant X-ray variability (by factors of 2–3) over the same timescales on which X-ray binaries are typically variable. Thus, over timescales of days, weeks, or months, we could expect to find evidence of variability. In contrast, GCs that have a high level of X-ray source multiplicity—i.e., in which the total flux is a composite consisting of comparable contributions from more than a few sources—should be almost constant, since it is unlikely that many of the independent components will have L declining or increasing simultaneously.

Angular resolvability.—At the distance of M104, 1″ corresponds to roughly 43 pc, larger than the radii of typical GCs. Most of the GC X-ray sources in our sample are located away from the aim point, in regions where the angular resolution may be as large as several arcseconds. If, therefore, some of the high-luminosity sources that we detect in bright GCs are actually composites, one or more of the individual components may consist of X-ray sources that have been ejected from the cluster. It is possible that a subset of such systems could be resolved by pointed observations with *Chandra's* High Resolution Camera.

5.5.3. *Giant Donors*

The observations we have described introduce a puzzle: the brightest blue GCs do not host X-ray sources, while dimmer blue GCs can house X-ray sources that are among the brightest in the galaxy. We therefore seek to understand

how they come to house extremely bright X-ray sources, while the brightest blue GCs do not appear to harbor X-ray sources, even X-ray sources that are ~10 times dimmer.

The thermal timescale scenario described above is likely to work in either young GCs, where the turnoff mass may be larger than 0.8 $M_{\odot}$, or in very massive GCs, in which the probability of a NS orbiting a blue straggler donor is larger. Similarly, multiplicity would seem to be most likely in large GCs with a high level of concentration and hence high central surface brightness.

The problem posed above would seem to require another solution. Presumably the donor stars are old and less massive than their NS companions. In such cases, the mass-transfer rate can be high if the donor is a giant. Such systems will exist in a GC only if a binary with total mass ~2.2 $M_{\odot}$ and a separation of ~50–1000 $R_{\odot}$ can survive. Some such binaries are considered "soft" in the GC environment, having less binding energy than the ambient kinetic energy of surrounding stars. Soft binaries have a good chance of being disrupted by a combination of distant interactions acting over time. Others of these binaries may not be formally "soft," but, since they are likely to be found near the cluster center, where close interactions are most frequent, they can also be disrupted. Both mechanisms are least efficient at destroying the binaries with large separations in less massive and less highly concentrated GCs. It may therefore be that the most X-ray luminous X-ray sources in the less bright blue GCs are accreting NSs with giant donors.

It would be difficult to obtain direct evidence that a given GC X-ray source, even in M31, was an example of a NS accreting mass from a giant donor. A useful form of indirect evidence could, however, come from GCs in our own Galaxy. Specifically, we could attempt to establish the existence, in some clusters with low present-day interaction probabilities, of relatively luminous ($L > 10^{32}$ ergs s^{-1}) cataclysmic variables with giant donors. Galactic GCs are near enough that the optical properties of the X-ray binaries can be studied with *HST*, allowing general features of the model to be checked. WDs accreting from giant donors, which cannot presently be studied in external galaxies, may be more common analogs of the accreting NSs with giant donors. As such, they could provide insight into the environments that nurture such systems.

6. CONCLUSION AND PROSPECTS

M104 may be one of the richest theaters in which to study X-ray emission from galaxies. In this paper we have focused on the emission from nonnuclear point sources, since the diffuse emission (see Fig. 2) that motivated this observation is the subject of a separate paper (W. R. Forman et al. 2003, in preparation; see also Delain et al. 2001), as is the nuclear source (Pellegrini et al. 2003a, 2003b).

The primary science results we have extracted so far from the 18.5 ks observation revolve around very soft sources and the connection between GCs and LMXBs. We find SSSs and QSSs inhabiting the bulge, disk, and halo of M104. At least four of the halo sources are very soft, likely as soft as the SSSs known in the Magellanic Clouds. These are good candidates for nuclear-burning accreting WDs. The discovery of these sources several kiloparsecs from the disk indicates that they are old systems, as should be expected for a subset of nuclear-burning WDs. In the halo, as well as in the disk and bulge, we also find very soft sources from which we have collected a small number of photons with energies above 1.1 keV. In some cases, photon statistics or the incomplete subtraction of background due to diffuse emission could be responsible for the harder photons. In most cases, however, the photons are likely to emanate from the system itself. Some SSSs and QSSs exhibiting this harder emission may be supernova remnants. Since, however, similar sources have been discovered in other galaxies, including M31, M101, NGC 4472, M51, and M83, and since we have discovered that many such SSSs and QSSs are variable on scales of months to years, it is likely that a large fraction is X-ray binaries. The harder photons from some SSS binaries may support a model in which the accretor is an intermediate-mass BH. We find a significant overdensity of SSSs within 1 kpc of the nucleus; future work will assess whether the overdensity is more pronounced for very soft sources than for other X-ray sources. If it is, this could be indicative of tidal disruptions of giants by the ~10^9 $M_{\odot}$ BH at the center of the Sombrero.

Regarding the LMXB/GC connection, we find that the fraction of bright X-ray sources in GCs is similar across galaxy types. We also find that those optically bright GCs that have X-ray sources house only bright X-ray sources. M104 provides another example of a galaxy in which X-ray sources are most likely to be found in red GCs; these clusters are presumably more metal-rich and younger than bluer clusters. Interestingly enough, this preference for redder clusters does not seem to be expressed by the brightest ($L_X > 10^{38}$ ergs s^{-1}) X-ray sources. Another interesting fact is that M104's optically brightest bluer clusters do not seem to house X-ray sources. We have sketched several models that can help to explain the trends seen in the data. Thermal timescale mass transfer can, in younger or more massive clusters, produce high-luminosity systems. Multiplicity can play a role in some massive clusters. Accretion by a NS from a giant could be occurring in some blue clusters that house X-ray bright sources.

6.1. *Prospects*

An ultimate goal of X-ray observations of galaxies is to better understand the natures of the point sources and the processes by which stellar populations create them. To that end, the fact that M104 has a large X-ray source population and is also well studied at other wavelengths supports the process of relating X-ray sources to their environments. Its nearly edge-on orientation is also a boon, because it provides a rare opportunity to explore regions out of the disk of a spiral galaxy with a minimum of projection effects.

The Sombrero is not only an important galaxy to study in its own right but also an important member of a set of external galaxies whose study will allow us to probe environmental effects on the creation, evolution, and properties of X-ray sources. Together with well-studied galaxies such as the Milky Way, M101, M83, and M81, it provides an example of an X-ray active spiral galaxy, supplementing information derived from the others because it is a bulge-dominated spiral and also because of its edge-on orientation. Consider the contrast with, e.g., M101, which is located at a comparable distance, but which is viewed nearly face-on. In M101 X-ray sources can be directly identified with disk stellar populations, but it is not possible to know which X-ray sources are actually located away from the

galactic plane. The combination of studies of M101 and M104 can therefore play complementary roles. Another key difference is that the central region of M101 is not known to house a central BH, while M104 harbors a central BH with $10^9\ M_\odot$. Already we know that the central concentration of X-ray sources in M101 is not as extreme as in M104 and that there are no bright SSSs and QSSs with 1 kpc of the nucleus (Di Stefano & Kong 2003a, 2003b). M31 does have a massive BH, but it is 30 times less massive than the Sombrero's; it will be important to compare the densities and properties of X-ray sources among these galaxies. Because the Sombrero provides a link between disk-dominated and bulge-dominated galaxies, it will also be productive to compare the X-ray source population of M104 with those of elliptical galaxies, such as NGC 4472 (Kundu et al. 2002; Maccarone et al. 2003), NGC 4697 (Sarazin et al. 2001), and NGC 1399 (Angelini, Lowenstein, & Mushotzky 2001).

6.1.1. *Future Observations*

Future observations can not only provide a profile of the X-ray source distribution in the Sombrero but can also answer a range of science questions related to variability and to the environmental influences that affect the formation and evolution of X-ray sources in galaxies.

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